

FDI Fell Algebras Have the Expected BCP Criterion

Run fa_banach_001, lane 19

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Source Problem

The source paper is Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257. After proving that a commutative unital C^* -algebra $C(K)$ has BCP/UBCP exactly when K has a countable π -basis, the authors ask for a noncommutative version:

$\{U_\alpha\}_{\alpha \in A}$ of K is called a π -basis (weak basis) for K if every nonempty open subset of K contains some U_α in $\mathcal{U}$ [37, Page 15].

In the setting of commutative C^* -algebra, [25, Theorem 2.1] can be reformulated as follows. Note that [25, Theorem 2.1] only deals with real continuous functions. The complex case is a direct consequence of the real case.

Corollary 3.9. *Let $\mathcal{A}$ be a unital commutative C^* -algebra. Then the followings are equivalent:*

- (1) $\mathcal{A}$ has the BCP;
- (2) the pure state space $\mathfrak{P}(\mathcal{A})$ has a countable π -basis;
- (3) $\mathcal{A}$ has the UBCP.

In particular, there exist commutative C^* -algebras containing no minimal projections but having the BCP. For example, the uniform norm closure of the linear span of Haar functions [24, p.150].

We conclude this section by the following question.

Question 3. *It will be interesting to find a non-commutative version of Corollary 3.9, that is, how to characterize those C^* -algebras $\mathcal{A}$ having BCP/UBCP.*

Definitions

Definition 1. *A C^* -algebra A has the ball-covering property, or BCP, if the unit sphere of A is covered by countably many open balls none of which contains the origin. It has the uniform ball-covering property, or UBCP, if the balls can be chosen with a common positive gap from the origin.*

Definition 2. *A C^* -algebra is FDI if all its irreducible representations are finite dimensional. Equivalently, every primitive quotient is a full matrix algebra $M_n(\mathbb{C})$, with n allowed to depend on the primitive ideal.*

Statement

Theorem 1 (FDI Fell criterion). *Let A be a Fell C^* -algebra all of whose irreducible representations are finite dimensional. Then*

$$A \text{ has BCP} \iff A \text{ has UBCP} \iff \text{Prim}(A) \text{ has a countable } \pi\text{-basis.}$$

Idea of the Proof

The earlier local compact-chart packet proves UBCP for any C^* -algebra whose primitive space has a countable π -basis of open charts on which every ambient element is a norm-continuous compact-operator field. The multiplier obstruction for general type-I algebras is that an element may restrict to a continuous-trace ideal only as a multiplier field, and multiplier fields over $\mathcal{K}(H)$ need not be norm-continuous.

For FDI Fell algebras this obstruction disappears. Fell spectra are locally Hausdorff. On a Hausdorff open patch, the corresponding ideal is Fell with Hausdorff spectrum, hence continuous trace. Since the irreducible representations are finite dimensional, the local fibres are matrix algebras $M_n(\mathbb{C})$. For matrix fibres, the multiplier algebra is just bounded norm-continuous matrix fields. Thus FDI Fell algebras satisfy the compact chart hypothesis, and the local chart theorem applies.

Imported Criteria and Standard Facts

We use the following criterion from the earlier lane-19 local compact-chart packet.

Proposition 1 (local compact-chart criterion). *Let B be a C^* -algebra. Suppose $\text{Prim}(B)$ has a countable π -basis (O_n) such that for each n :*

1. *the ideal B_{O_n} is isomorphic to $C_0(O_n, \mathcal{K}(H_n))$ for a separable nonzero Hilbert space H_n ;*
2. *for every $b \in B$, the image of b on O_n is represented by a bounded norm-continuous function $O_n \rightarrow \mathcal{K}(H_n)$, recovering the primitive quotient norms.*

Then B has UBCP.

We also use two standard Fell-algebra facts. First, the spectrum of a Fell algebra is locally Hausdorff. Second, a Fell algebra with Hausdorff spectrum has continuous trace. The first appears, for example, as Lemma 3.2 of Clark–an Huef–Raeburn [2], and the second is the classical Dixmier criterion recalled in their introduction.

Proof

Lemma 1 (FDI Fell algebras have compact-norming charts). *Let A be an FDI Fell C^* -algebra. If $\text{Prim}(A)$ has a countable π -basis, then A satisfies the hypotheses of Proposition 1.*

Proof. Write $X = \text{Prim}(A)$, and let $\mathcal{P} = \{P_m : m \geq 1\}$ be a countable π -basis for X .

We first refine $\mathcal{P}$ to Hausdorff continuous-trace trivialization patches. Let $U \subset X$ be nonempty open. Choose $x \in U$. Since A is Fell, X is locally Hausdorff, so there is an open Hausdorff neighbourhood V of x with $V \subset U$ after shrinking inside U . The ideal A_V is again Fell, and its spectrum is the Hausdorff space V . Hence A_V has continuous trace.

By local triviality of continuous-trace algebras, there is a nonempty open set $W \subset V$ such that

$$A_W \cong C_0(W, \mathcal{K}(H))$$

for some Hilbert space H . Because A is FDI, every irreducible representation over W is finite dimensional; therefore H is finite dimensional on this trivialization. Since $\mathcal{P}$ is a π -basis for X , there is $P_m \in \mathcal{P}$ with

$$\emptyset \neq P_m \subset W \subset U.$$

Restricting the above trivialization gives

$$A_{P_m} \cong C_0(P_m, M_d(\mathbb{C}))$$

for $d = \dim H < \infty$. Thus the subfamily of all P_m 's that occur in such finite-dimensional continuous-trace trivializations is still a countable π -basis of X .

Fix one such chart $O = P_m$, and identify

$$A_O \cong C_0(O, M_d(\mathbb{C})).$$

Let $a \in A$. The canonical action of A on the ideal A_O gives a multiplier

$$\phi_O(a) \in \mathcal{M}(A_O).$$

Since $M_d(\mathbb{C})$ is unital and finite dimensional,

$$\mathcal{M}(C_0(O, M_d(\mathbb{C}))) \cong C_b(O, M_d(\mathbb{C})),$$

with the usual norm-continuous bounded sections. Let

$$\hat{a}_O : O \rightarrow M_d(\mathbb{C})$$

be the section corresponding to $\phi_O(a)$. This section is bounded and norm-continuous.

For $P \in O$, evaluation at P gives the irreducible representation of the ideal A_O , and the multiplier action is exactly the restriction of the irreducible representation of A with kernel P . Hence

$$\|a + P\| = \|\hat{a}_O(P)\|.$$

Since $M_d(\mathbb{C}) = \mathcal{K}(\mathbb{C}^d)$, the chart is a compact-norming chart in the sense of Proposition 1. The countable family constructed above therefore verifies all hypotheses of Proposition 1. $\square$

Proof of Theorem 1. If $\text{Prim}(A)$ has a countable π -basis, then Lemma 1 verifies the compact-norming chart hypothesis for A . Proposition 1 gives UBCP.

The implication $\text{UBCP} \Rightarrow \text{BCP}$ is immediate. Conversely, if A has BCP, the primitive π -base obstruction from the earlier lane-19 packet gives a countable π -basis for $\text{Prim}(A)$. This proves the equivalence. $\square$

Scope and Limitations

This theorem is a positive non-Hausdorff Fell/type-I result, but it is not the full Fell theorem. The finite-dimensionality of irreducible representations is doing real work: it makes

$$\mathcal{M}(C_0(O, M_d(\mathbb{C}))) = C_b(O, M_d(\mathbb{C}))$$

a norm-continuous matrix-field algebra. For infinite-dimensional compact fibres $C_0(O, \mathcal{K}(H))$, multiplier fields live in a strict-continuous $B(H)$ -bundle, and pointwise compactness or Fell local rank-one behaviour does not automatically give norm-continuous compact sections. That strict-versus-norm gap is the remaining obstacle for Fell algebras with infinite-dimensional separable irreducible representations.

Novelty and Search Bounds

The local indexes were searched for arXiv:2311.14257, FDI, subhomogeneous Fell algebras, finite-dimensional irreducible representations, local compact charts, BCP, and UBCP. Web searches on June 17, 2026 for Fell algebras, locally Hausdorff spectrum, continuous trace patches, and arXiv:1207.2540 located the Clark–an Huef–Raeburn Fell reference [2], but no BCP criterion for FDI Fell algebras. The result is therefore recorded as an agent-produced partial theorem, subject to human literature review.

Human Review Focus

The proof is short because it delegates the covering construction to the local compact-chart packet. The main review points are:

1. the refinement of a countable π -basis to Hausdorff continuous-trace trivialization patches;
2. the claim that finite-dimensional fibres turn local multiplier sections into norm-continuous matrix-valued functions;
3. the norm-recovery identity $\|a + P\| = \|\widehat{a}_O(P)\|$ for ambient elements acting as multipliers of the local ideal.

References

- [1] Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257, 2023.
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- [3] J. Dixmier, *C*-Algebras*, North-Holland, 1977.
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- [5] Run `fa_banach_001`, lane 19, *A Local Compact-Chart Criterion for C*-Algebras with UBCP*, solution packet `2311.14257_local_compact_chart_bcp_criterion`, 2026.
- [6] Run `fa_banach_001`, lane 19, *A Primitive-Space π -Base Obstruction for C*-Algebras with BCP*, solution packet `2311.14257_primitive_pi_base_obstruction`, 2026.